RFID

While it is not technically an application of embedded systems, the use of RFID is going to be boosted quite a bit as it now RFID technology has reached the potential to now become an integral part in embedded-system design. Traditionally used in inventory management, developments in RFID technology along with high-speed, long-range readers now allow embedded-system designers to incorporate with much ease features such as simplified payments, counterfeit prevention, medical authentication, access control, dynamic pricing, remote asset tracking, and product histories. By incorporating the reader in an even deeper level within the product or system, Embedded RFID applications add local data-gathering features which serve to improve greatly the originally slated functions of the product. Embedded-RFID applications are found in a lot of countries in restaurants, prisons, shopping outlets, casinos, toll roads, factories, and vehicles.

One of the initial uses of RFID was by the US military (who seem to have always have been on top of every technology out now) during World War II. It was used to distinguish between enemy and friendly fire aircraft. By the 1970s and through the 1980s it was commercially applied to be able to track and identify items within a location. The problem here was that there was no standardisation of the software- with each developer presumably looking to get one-up over the other by using their own unique methods of communications. Because of this loss of brotherhood among these developers, the industry fell apart. There was very slow adoption, probably because everyone was wasting their time trying to figure out the nuances of every new RFID communication device with its unique signature, and because of this RFID was just considered to be just another one of those new technologies which was hyped up so much that when it fell all people could do was to feel embarrassed they were ever part of it. Today, though, things have changed for the better. Developers realised where they had gone wrong in the past, and have been working on getting RFID back to the pedestal it once was on.

The Lexus RX330 had problems with the embedded anti-lock brake system